

Marc Recio

Data scientist · plant genomics

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SUMMARY

Data scientist working in plant genomics. I came from the wet lab — in vitro culture, PCR, sequencing — and now build the pipelines that read what those experiments produce, designed so a scientist can audit the calls the machine makes. Current work pairs large language models with deterministic, tested code so that every reproducible decision is made by the code and never by the model. Classical machine learning, databases and the full data lifecycle; deep learning and reinforcement learning in progress. Fluent English, and experience in international, research-driven environments.

SELECTED WORK

Evidence-graded gene curation from the literature

2025 – present

MSc data-science thesis · with the TomsBioLab research group

- Pipeline that turns grapevine-genomics papers into evidence-graded catalogue rows: language models read, deterministic code makes every call that has to be reproducible.
- The model never assigns the evidence level. It answers a fixed set of yes/no questions and tested code maps those booleans onto the published 6-level scale — so a level can be re-derived, audited and unit-tested, which an opinion cannot.
- **98.7 % level accuracy** against a hand-curated gold set — 74 of 75 genes across 9 scorable papers. Providers sit behind one interface, so a lab can run the whole pipeline on its own model.
- Findings are held for the January defence; the engineering is documented in the portfolio.

High-throughput screen for heat-stress protection in pollen

2024

MSc plant-breeding thesis (COMAV, UPV) · graded 10/10

- Screened 800 compounds for protection of rapeseed (*Brassica napus* DH4079) pollen against a heat shock that collapses germination from ≈61 % to below 2 %; returned 8 candidate hits as a shortlist for follow-up.
- Designed the assay, ran the statistics behind the results, and reported the negative controls — DMSO protects at mild heat (32 °C) but not at 38 °C.

Published dataset · scraper and data contract

2025

Zenodo, DOI 10.5281/zenodo.17578619 · CC BY-NC-SA 4.0

- 1,000 films × ≈25 fields in a single CSV. Per-field error handling so one missing element never aborts a record, explicit waits instead of brittle sleeps, and resumable output that survives interruption.

EXPERIENCE

Research staff

May 2024 – Dec 2024

COMAV — Institute for Conservation and Improvement of Valencian Agrobiodiversity (UPV), Valencia

- Developed a protocol for obtaining transgenic rapeseed and ran in vitro microspore cultures (gametophytic development and androgenesis).
- PCR and sequencing for genetic analysis, plus histological analysis of salinity impact on reproductive development of *Cakile maritima*.
- Designed and carried out the statistical analyses behind the experimental results.

Laboratory technician (internship)

Jun 2023 – Nov 2023

Semillas Fitó, Cabrera de Mar, Barcelona

- In vitro procedures producing double haploids in pepper, cucumber and watermelon.
- DNA-extraction protocols in the molecular-biology department.

SKILLS

Machine learning	Supervised and unsupervised methods, model evaluation on imbalanced data (F1-macro, class weighting), dimensionality reduction (PCA, t-SNE, UMAP).
LLM engineering	Structured extraction, prompt design against a gold set, provider-agnostic interfaces, rule-based post-processing of model output.
Data & databases	SQL, NoSQL, schema design, data modelling, ETL, cleaning and preprocessing.
Programming	Python, R, SQL. Modular packages, unit tests, command-line and API entry points.
Collection & visualization	Web scraping, APIs, exploratory analysis, reporting, Power BI.
Domain	Plant genomics and breeding, molecular biology, experimental design, statistical analysis.
Languages	Spanish (native), Catalan (native), English (fluent — B2 First certified), French (A2).

EDUCATION

MSc in Data Science	Feb 2025 – expected 2026
Universitat Oberta de Catalunya (UOC) · top grade in Machine Learning	
MSc in Plant Genetic Breeding	Sep 2021 – Sep 2024
Universitat Politècnica de València (COMAV) · thesis graded 10/10	
BSc in Food Science and Technology	Sep 2017 – Sep 2021
Universitat Politècnica de València	

CERTIFICATES

• CS50's Introduction to Programming with Python — HarvardX (edX)	• Cambridge English: B2 First — Cambridge Assessment English
• Data Science: R Basics — HarvardX PH125.1x (edX)	• Valencian language, level C1 — JQCV